

The Purple-Pipe Loop

Ladera Ranch's runoff-recycling system, mapped and audited · LEHRP · September 2026

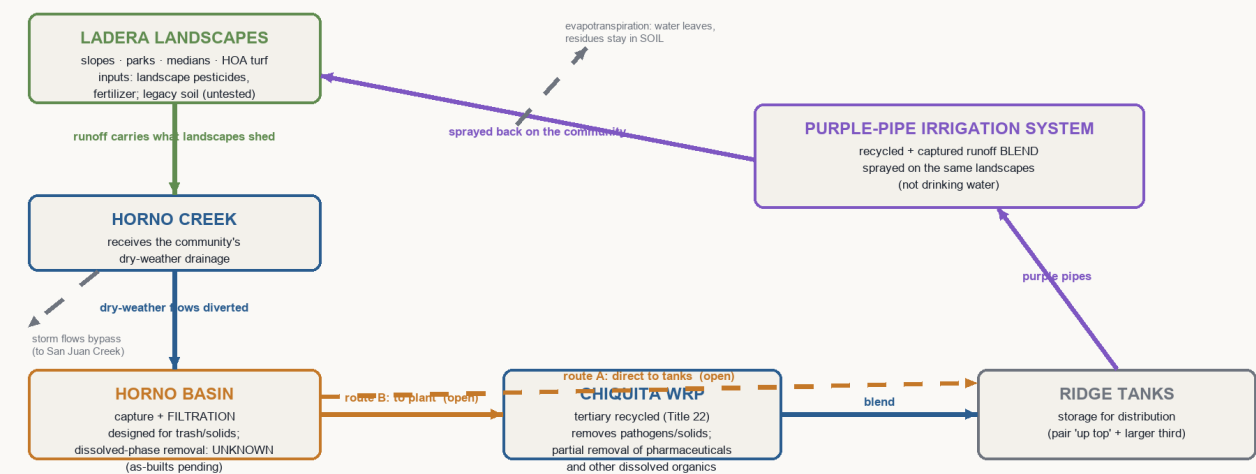
This is an independent research and data-organization project. It does not provide medical advice and does not establish that any party caused any illness. No contaminant is asserted to be present in any water or soil discussed here; the relevant streams have not been publicly tested for the constituents examined, which is itself a central finding. Every claim carries a statement class. Geographic and temporal overlap does not establish exposure or causation.

1 · The loop is real — verified (A2)

The water district's own materials establish the geometry. Its recycled-water page: the district "collects urban return (runoff) water from Oso Creek, Dove Creek, and **Horno Creek** to **blend with the recycled water** as an additional source **for irrigation**." Its May 2012 newsletter: the "**Horno Basin**... designed to capture, treat and deliver millions of gallons of runoff water **for local irrigation use**." The delivery network is the purple-pipe (recycled, non-potable) system irrigating Ladera's slopes, parks, medians and common-area turf. Water that drains off the community's landscapes is captured at the bottom of the watershed and a portion is sprayed back onto those same landscapes. Design intent, stated by the operator.

The Ladera runoff-recycling loop, drawn honestly

Solid arrows = documented (A2, water district sources). Dashed = plausible pathway, unmeasured. Purple = the community's purple-pipe (recycled/non-potable) irrigation system.



AUDIT RESULT

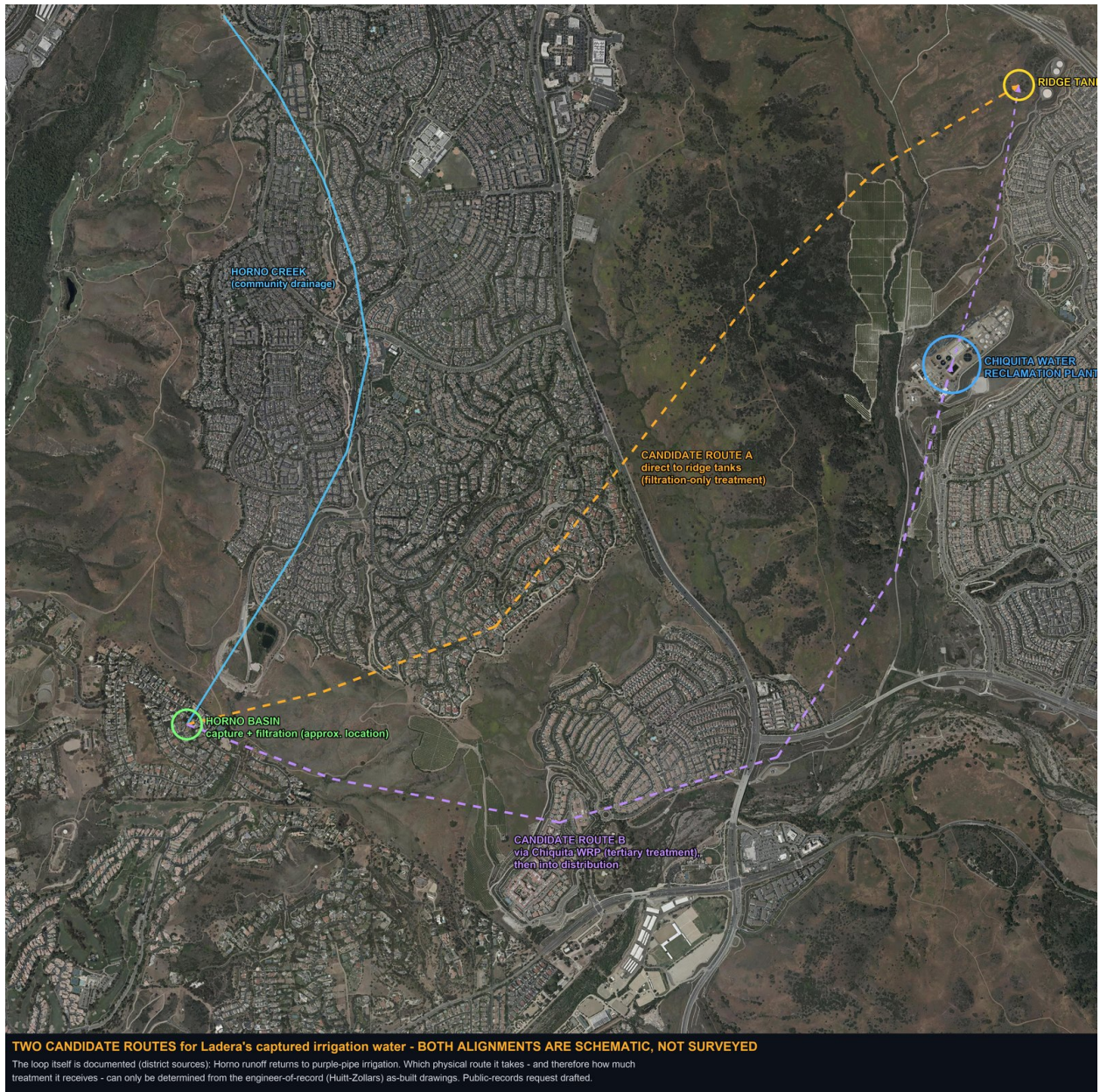
Water-phase 'compounding' is capped by the loop's leaks: +2% to +28% enrichment even with aggressive assumptions - runaway buildup in the WATER fails the math. What survives audit: REDISTRIBUTION (drainage constituents re-sprayed community-wide), LINEAR SOIL ACCUMULATION (residues left behind year after year), and the MONITORING GAP - no public arsenic/pesticide/pharmaceutical data for the captured-runoff stream. All three are testable with inexpensive samples.

LEHRP - statement classes marked - no contamination is asserted; the loop geometry is documented, its chemistry is unmeasured

The loop, drawn honestly: documented flows solid, open questions dashed.

1b · Two candidate routes — settled only by the drawings

Both routes end in the same purple pipes, so the loop is confirmed either way; the route determines how much treatment the captured runoff receives first. **Route A** (direct to the ridge tanks): basin-level screening, settling, filtration and disinfection only — sanitary and clear, but dissolved metals, dissolved pesticides and pharmaceuticals pass through. **Route B** (via the Chiquita Water Reclamation Plant): full tertiary treatment is possible — genuinely more — though tertiary treatment is not engineered for dissolved arsenic either, and some systems blend runoff in downstream of treatment, adding nothing. The alignments below are SCHEMATIC, NOT SURVEYED: the actual route exists in the engineer-of-record (Huitt-Zollars) as-built drawings, obtainable by public-records request from the district (draft prepared).



County 2025 imagery. Amber = Route A (filtration only). Purple = Route B (via Chiquita WRP). Blue = Horno Creek. Neither alignment is asserted.

2 · The audit — what fails, what survives

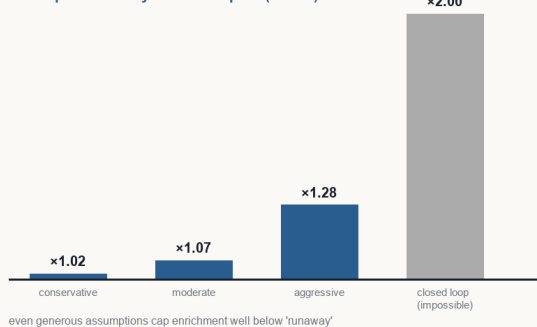
FAILS: "compounds" in the water. The loop is partial: storm flows bypass capture, evapotranspiration removes water, captured runoff blends into a much larger supply. A steady-state model caps water-phase enrichment at $\times 1.02$ – $\times 1.28$ even with aggressive assumptions (a fully closed loop, which is impossible, would only reach $\times 2.00$). Runaway concentration in the water is not physically supported.

SURVIVES — redistribution: whatever the drainage carries gets pumped uphill and sprayed across every purple-pipe-irrigated surface in the community. **SURVIVES — soil accumulation:** irrigation water evaporates; persistent constituents (metals including arsenic, persistent pesticide residues) stay in landscape soils and accumulate roughly linearly with years of application — two-plus decades so far. Magnitude unmeasured.

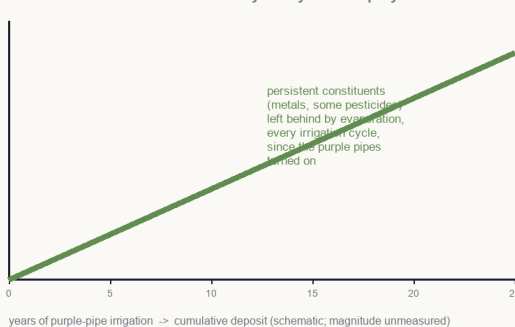
SURVIVES — the monitoring gap: the drinking-water system is tested and separate; the captured-runoff stream, basin sediment and irrigated soils have no public testing record for pesticides, pharmaceuticals or legacy arsenic.

Why the risk lives in the SOIL, not the water

Water-phase steady-state multiplier (model)



Soil reservoir: accumulates linearly with years of spray



3 · The chemicals, one at a time

Constituent	Enters the loop how	Evidence status
Landscape pesticides (fipronil, pyrethroids)	Applied on landscaping; wash off in runoff — the classic So-Cal urban-runoff signal	Regional presence: B1. Levels in THIS loop: untested
Pharmaceuticals / trace organics	Partially survive tertiary treatment; enter via the recycled blend	Phenomenon: B1. This blend: not publicly reported
Arsenic (legacy, 1907–1912 era)	IF present in drainage soils (unmeasured), mobilizes in runoff; filtration does not remove the dissolved phase	Hypothesis — no soil/runoff arsenic data exists here; the historical program on this land is A1-documented

VERDICT

Not supported as stated: "one big ongoing toxic loop that compounds" — water-phase compounding fails mass balance. **Supported, restated defensibly:** Ladera Ranch operates a documented capture-and-respray loop through its purple-pipe system. That loop redistributes whatever the community's drainage carries onto its landscapes, deposits persistent constituents into landscape soils year after year, and does so with no public monitoring of the captured-runoff stream — on land where a federal arsenical dipping program operated a century ago and where soil has never been tested for it. The reported pattern warrants investigation; current data are insufficient to determine whether any constituent is present at levels of concern.

4 · What would settle it — four inexpensive samples

1) Irrigation water at a purple-pipe sprinkler head (arsenic + speciation, pesticide panel, CEC screen) — what the loop delivers today; the cheapest, highest-value test. 2) Horno Basin sediment core — what the watershed has been shedding, time-resolved. 3) Long-irrigated turf/slope soil vs never-irrigated control — whether two decades of spray built a measurable deposit. 4) Blend-point water at tank/plant (pending as-builts) — which route, and what treatment it actually receives.

Statement classes: Verified fact · Official statement · Scientific finding (B1) · Model estimate · Inference · Hypothesis · Missing evidence. Sources: district recycled-water page and May 2012 newsletter (archived, SHA-256), county aerial imagery 2025, regional urban-runoff and water-recycling literature. LEHRP · corpus timestamped under version control.